Samuel Maxwell Obeng Avornyoh

Software & AI Engineer

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SUMMARY

- Software and AI engineer who designs and ships complete systems: Rust and Go services, PostgreSQL data models, on-chain settlement, encrypted storage and retrieval-grounded AI assistants.
- Built BitMomo (crypto payments for Ghanaian merchants, on testnet), ProxyHubb (Solana-escrowed proxy marketplace, live), CipherVault (encrypted secrets backup, v1.0.32) and CREG (consensus package registry, public alpha).
- Documents architecture, release evidence and limits clearly; comfortable owning a product from design through cloud deployment, alone or in small teams.

EXPERIENCE

Blockchain / Gaming / AI Project Developer | Independent projects — May 2025 - Present

- Building BitMomo, a crypto payment platform for Ghanaian merchants — 13 Rust crates, four Next.js portals and a GCP testnet deployment with release evidence.
- Built ProxyHubb, a proxy marketplace with a Go control plane, Rust edge network and Solana escrow; marketplace and docs are live.
- Released CipherVault v1.0.32, a Rust tool for client-side encrypted backup and recovery of developer secrets.
- Built CREG, a validator-consensus package registry (public alpha) with PBFT, IPFS and Sepolia contracts.

Full-Stack Developer | TonyCold Store Management System — May 2026 - Present

- Developed a business management system for a cold store dealing in frozen meat, fish and related products.
- Designed modules for product records, inventory batches, sales/POS operations, stock movement, expense tracking and reporting.
- Improved operational visibility by organizing product categories, pricing, sales summaries and admin workflows.
- Made profit reporting accurate by costing sales from inventory batches and separating wholesale profit from retail revenue.

National Service Personnel / IT Technician | Sefwiman Community Bank — Mar 2026 - Sep 2026

- Maintained networking systems and handled troubleshooting.
- Supported bank operations and cybersecurity.

National Service Personnel / AI & Data Engineer | Really Great Tech — Dec 2025 - Mar 2026

- Built and maintained an AI/data training repository covering analytics, SQL, dashboards, supervised ML, deployment, MLOps, LLM fundamentals, LangChain, RAG and model evaluation.
- Designed weekly workflows and milestone projects using Git branches and pull requests, Python, Jupyter, scikit-learn, SQL and Looker Studio.
- Developed learning materials and documentation for business-insight, ML-microservice and telecom-policy-assistant use cases.

Machine Learning Intern | Makersplace — Sep 2023 - Jan 2024

- Created a web-based generative AI assistant that supports robotic facilitators in course delivery, grounded in a facilitator guidebook.
- Built charts that communicate AI-agent activity, performance and decisions to non-technical stakeholders.
- Developed machine-learning models for financial-market forecasting and trading-strategy research.
- Integrated AI agents with Telegram and WhatsApp using FastAPI and messaging libraries.

SELECTED PROJECTS

BitMomo — Testnet · bitmomo.app — Feb 2026 - Present

- Designed a payment flow that credits merchants only after independent on-chain receipt and finality verification, with a ledger, reservations, outbox and approval-bound payouts.
- Built 13 Rust crates (Axum API, worker, chain monitor, settlement), four Next.js portals and two Expo apps; deployed to Google Cloud with Terraform, Cloud SQL and row-level security.

ProxyHubb — Live · proxyhubb.com — Jan 2026 - Present

- Built a Go control plane, Rust edge gateway and provider daemon (QUIC/Yamux, SOCKS5/CONNECT) and Anchor escrow programs that settle metered sessions on Solana devnet.
- Shipped a Next.js marketplace, Chrome extension, Node/Python/Go SDKs and a streaming account assistant with an evaluation gate.

CipherVault — Released · vault.cipherv.online Sep 2026 - Present

- Wrote a Rust workspace for client-side encrypted backups: XChaCha20-Poly1305, HKDF-keyed FastCDC chunking, Ed25519-signed snapshots and M-of-N guardian recovery.
- Added safe secret injection for processes and AI agents with exact output scrubbing and a hash-chained audit log; released v1.0.32.

CREG — Chain Registry — Public alpha · contribute.cregnet.dev Mar 2026 - Present

- Built a validator node in Rust (libp2p, PBFT, RocksDB), a static/sandbox/deep-scan package pipeline and Sepolia staking, registry and governance contracts.
- Shipped a fail-closed creg CLI with npm/pip/cargo shims, SBOM generation and signed v0.4.8-testnet binaries.

NeuroBench (NeuroState) — Prototype · github.com/samuel-1-avson/Neurostate Dec 2025 - Mar 2026

- Tauri IDE (Rust + SolidJS) for FSM design and embedded code generation; one of 50 winners of Google DeepMind's Gemini 3 hackathon.

PosAI — In development Oct 2026 - Present

- Lead backend contributor (team project): tenant-isolated TypeScript services, capability registry, deterministic compiler and leased job workers.

SMS AI Ghana — MVP · github.com/samuel-1-avson/sms-ai Oct 2026 - Present

- Go service delivering grounded AI answers over SMS/USSD with signed webhooks, idempotency and multipart coalescing; MVP tested against mock gateways.

More — Agentic QA Platform, Generator Sound Health Monitor, healthcare no-show prediction and others — at samuel-seven-gamma.vercel.app/work.

SKILLS

Languages: Rust, TypeScript, Go, Python, SQL, Solidity, C/C++, Bash
Backend & systems: Axum / Tokio, Tonic gRPC, FastAPI, Express, tRPC, PostgreSQL, Redis, RabbitMQ, SQLite, libp2p
Blockchain & payments: Solana / Anchor, EVM / Sepolia / Base, Tron USDT, HD wallets (BIP-44 / SLIP-10), PBFT consensus, IPFS, Groth16 / Circom, Ledger & payout design
Security: XChaCha20-Poly1305, HKDF / Ed25519, M-of-N recovery, Row-level security, Supply-chain checks (SBOM), Secret scanning, Threat modelling
AI & machine learning: LLM applications (Gemini, Llama), Retrieval-augmented generation, Chroma, PyTorch, TensorFlow, scikit-learn, MLflow, LangChain, Evaluation gates
Frontend & apps: React, Next.js, SolidJS, Tauri, Expo / React Native, Tailwind CSS, Playwright
Cloud & operations: Google Cloud, Terraform, Docker, Kubernetes, nginx / Caddy, Cloudflare, GitHub Actions, Prometheus / Grafana / OpenTelemetry

EDUCATION

BSc Computer Science and Engineering | University of Mines and Technology, Tarkwa, Ghana Nov 2025

Second Class Upper. Coursework: Artificial Intelligence, Data Structures and Algorithms, Database Systems, Data Science Fundamentals, Probability and Statistics, Embedded System Design, Linear Algebra, Robotics.

RECOGNITION

Winner — Vibe Code with Gemini 3 Pro in AI Studio — Google DeepMind · Kaggle. One of 50 winning projects, for NeuroState, an AI-powered embedded systems IDE.

Languages: English, Asante Twi · Revised Nov 2026